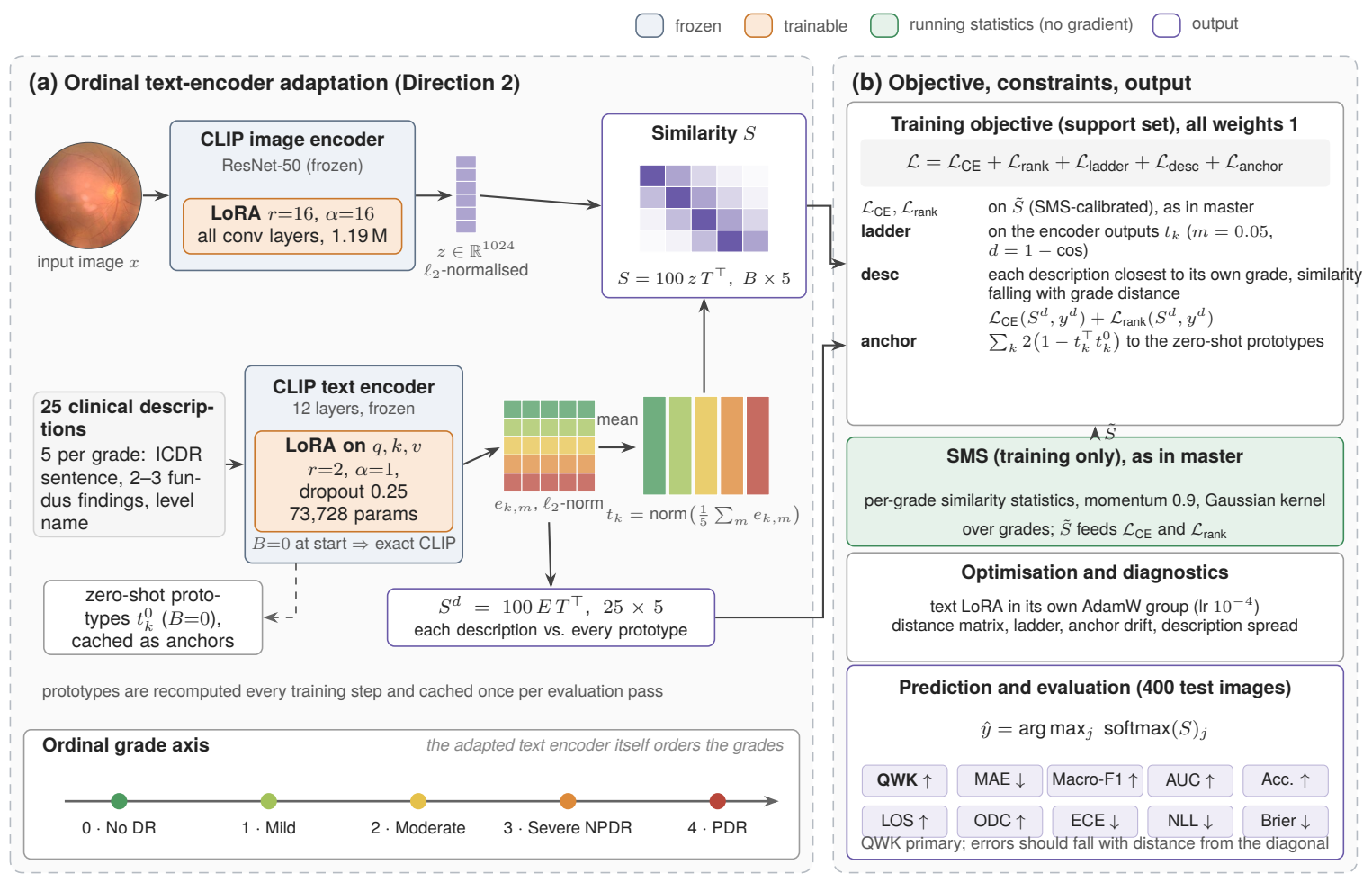
frozen
trainable
running statistics (no gradient)
output
(a) Ordinal text-encoder adaptation (Direction 2)
input image x
CLIP image encoder
ResNet-50 (frozen)
LoRA r=16, α=16
all conv layers, 1.19 M
$z \in \mathbb{R}^{1024}$
ℓ_2-normalised
Similarity S
$S = 100\, z\, T^\top,\ B \times 5$
25 clinical descriptions
5 per grade: ICDR sentence, 2–3 fundus findings, level name
CLIP text encoder
12 layers, frozen
LoRA on q, k, v
r=2, α=1,
dropout 0.25
73,728 params
B=0 at start ⇒ exact CLIP
$e_{k,m}$, ℓ_2-norm
mean
$t_k = \mathrm{norm}\left(\frac{1}{5}\sum_m e_{k,m}\right)$
zero-shot prototypes t_k^0 (B=0), cached as anchors
$S^d = 100\, E\, T^\top,\ 25 \times 5$
each description vs. every prototype
prototypes are recomputed every training step and cached once per evaluation pass
Ordinal grade axis
the adapted text encoder itself orders the grades
0 · No DR
1 · Mild
2 · Moderate
3 · Severe NPDR
4 · PDR
(b) Objective, constraints, output
Training objective (support set), all weights 1
$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{ladder}} + \mathcal{L}_{\text{desc}} + \mathcal{L}_{\text{anchor}}$
$\mathcal{L}_{\text{CE}}$, $\mathcal{L}_{\text{rank}}$ on $\tilde{S}$ (SMS-calibrated), as in master
ladder on the encoder outputs t_k ($m = 0.05$, $d = 1 - \cos$)
desc each description closest to its own grade, similarity falling with grade distance $\mathcal{L}_{\text{CE}}(S^d, y^d) + \mathcal{L}_{\text{rank}}(S^d, y^d)$
anchor $\sum_k 2\left(1 - t_k^\top t_k^0\right)$ to the zero-shot prototypes
$\tilde{S}$
SMS (training only), as in master
per-grade similarity statistics, momentum 0.9, Gaussian kernel over grades; $\tilde{S}$ feeds $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{rank}}$
Optimisation and diagnostics
text LoRA in its own AdamW group (lr 10^{-4})
distance matrix, ladder, anchor drift, description spread
Prediction and evaluation (400 test images)
$\hat{y} = \arg\max_j\ \mathrm{softmax}(S)_j$
QWK ↑
MAE ↓
Macro-F1 ↑
AUC ↑
Acc. ↑
LOS ↑
ODC ↑
ECE ↓
NLL ↓
Brier ↓
QWK primary; errors should fall with distance from the diagonal